EBN 111 Practical 1: Pre-practical Assignment

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Name and Surname: ______________________________

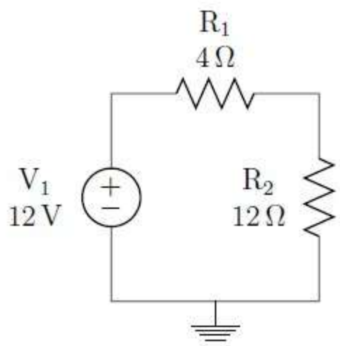
Student Number: ______________________________

Instructions

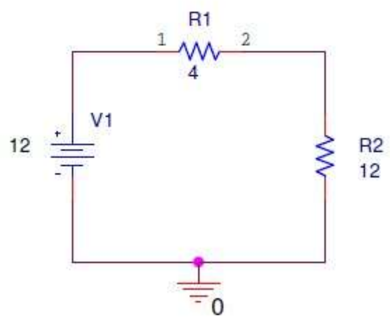
1. Print the pre-practical assignment
2. Fill in your student details.
3. Complete all questions found in this assignment.
4. Attach printed graphs to the report.
5. **Bring the completed assignment to practical 1.**

Question 1

1. Capture the common circuit schematic of Figure 1 in the Orcad Capture CIS program as shown in Figure 1b and simulate the circuit. You can either use a Bias Point Simulation or a Time Domain Simulation, whichever you find more convenient. Answer the questions that follow.



(a) Circuit diagram



(b) Simulated circuit diagram

Figure 1: Circuit diagrams for Question 1

(a) What is the total current in the circuit? ____________ A

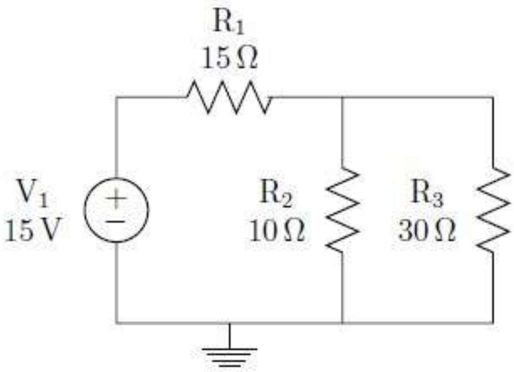
(b) What is the voltage at point 1 of R1? ____________ V

(c) What is the voltage at point 2 of R1? ____________ V

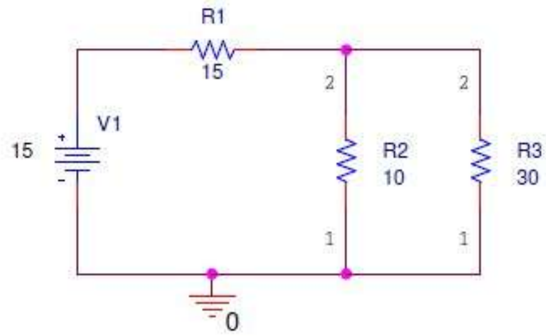
(d) What is the voltage difference across R1? ____________ V

Question 2

2. Capture the common circuit schematic of Figure 2a in the OrCAD Capture CIS program as shown in Figure 2b and simulate the circuit.



(a) Circuit diagram



(b) Simulated circuit diagram

Figure 2: Circuit diagrams for Question 2

(a) Perform a Bias Point or Time Domain simulation for Figure 2 and answer these questions.

i. What is the total current in the circuit? ______________ A

ii. What is the current at point 2 of R2? ______________ A

iii. What is the current at point 2 of R3? ______________ A

(b) Complete a DC Sweep simulation for Figure 2 sweeping the voltage source V1 from 0V to 10V with steps of 1V. (Be creative and google DC sweep if you want to know how to perform a DC sweep or use Help in OrCAD)

i. Attach a single graph of a DC Sweep of V1 from 0 to 10 V measuring the current through R1, current through R2 and current through R3.

ii. Attach a single graph of a DC Sweep of V1 from 0 to 10 V measuring the voltage across R1, the voltage across R2 and the voltage across R3.

Note: Each graph should have three separate lines, one line for each measurement. For each measurement probe that is added in OrCAD Capture, a new line is plotted in the SCHEMATIC window. Each line in the SCHEMATIC window will have the same color as the color of the probe in OrCAD Capture. The legend for each line is shown on the left, just below the x-axis of the graph.